

Quantum Mechanics-1 (Assignment-2)

(Dated: September 21, 2024)

1. The Radial part of the Hydrogen atom satisfies

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR(r)}{dr} \right) + \frac{2\mu}{\hbar^2} \left(E + \frac{ze^2}{r} - \frac{l(l+1)\hbar^2}{2\mu r^2} \right) R(r) = 0$$

Solve $R(r)$ in the limit $r \rightarrow \infty$ and find energy eigenvalues.

2. Given a supersymmetric potential $W(x)$, show that only one of the pairing Hamiltonians, i.e., either H_1 or H_2 can have zero energy bound state such that supersymmetry is unbroken.
3. A superpotential is given as $W(x) = 2\Theta(x - x_0) - 1$. The function $\Theta(x - x_0)$ represents a step function at $x = x_0$; i.e.,

$$\Theta(x - x_0) = \begin{cases} 0 & \text{for } x < x_0; \\ \frac{1}{2} & \text{for } x = x_0; \\ 1 & \text{for } x > x_0. \end{cases}$$

- (a) Derive the two pairing potential H_1 and H_2 .
- (b) Show that this superpotential leads to unbroken supersymmetry.
- (c) Determine the eigenfunction corresponding to the zero energy ground state of H_1 .
- (d) From the partner potential V_2 , determine the number of bound states supported by an attractive Delta function potential.
4. For superpotential $W(x, a) = a \tanh x$,
- (a) Show that it correspond to shape invariant potential.
- (b) Determine the complete bound state spectrum of $V_1(x, a)$ and first two eigenfunctions.
- (c) Determine the upper bound and lower bound of a such that there exists only 3 bound states.
- (d) Find out the condition on a such that V_1 and V_2 become reflectionless potential.
5. Given a superpotential $W(r, l) = \frac{a}{l} - \frac{l}{a} \left[\frac{1+e^{-r/a}}{1-e^{-r/a}} \right]$
- (a) Calculate the partner potentials and show that it corresponds to shape invariant potential.
- (b) Determine the complete bound state spectrum and two lowest eigenfunctions.
6. Consider the Eckart superpotential $W = -A \coth r + \frac{B}{A}$.
- (a) Show that it is shape invariant and find the condition on A and B , such that the supersymmetry is unbroken.
- (b) Determine relative values of parameters A and B such that the Eckart potential holds at least three bound states.